

Laser Radius Sensor Selection

EVKA Spherical Positioning System — Requirement Study

EVKA Position Project

2026-07-01

Requirement & Scope

Only the radius source changes. Two rotary encoders (Autonics E40S6, 20 000 PPR) keep measuring θ and ϕ — unchanged. The draw-wire is replaced by a laser distance reading for r .

Fixed requirement (confirmed with project owner)

- Range: ≤ 40 m
- Laser device accuracy: ≤ 10 mm (target sub-cm) across full range
- This is the *device datasheet spec* — a selection filter, not total XYZ accuracy

Version A — detachable handheld, docked on the phi head; quick-disconnect for field use.

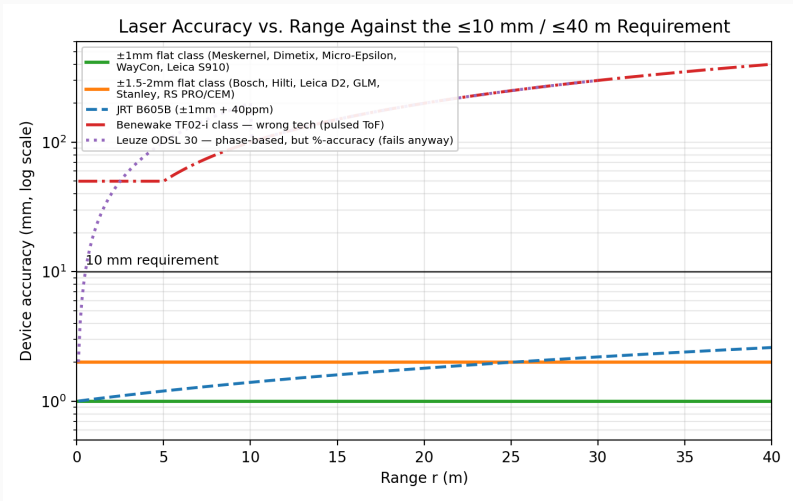
Version B — permanently mounted fixed module; automation-first, no handling.

Why Phase-Shift, Not Pulsed Time-of-Flight

Pulsed ToF (Benewake TF-series, DFRobot) — times a light pulse directly → ± 5 cm to $\pm 1\%$ of reading. **Fails the spec**, excluded.

Phase-shift / AMCW (Leica, Meskernel, Dimetix) — measures beam phase → **flat** $\pm 1\text{--}2$ mm at any range. Every **certified** device is phase-shift; Bosch & JRT are **flagged** — principle unconfirmed (2026-07-08).

Accuracy vs. Range — ≤ 10 mm Requirement



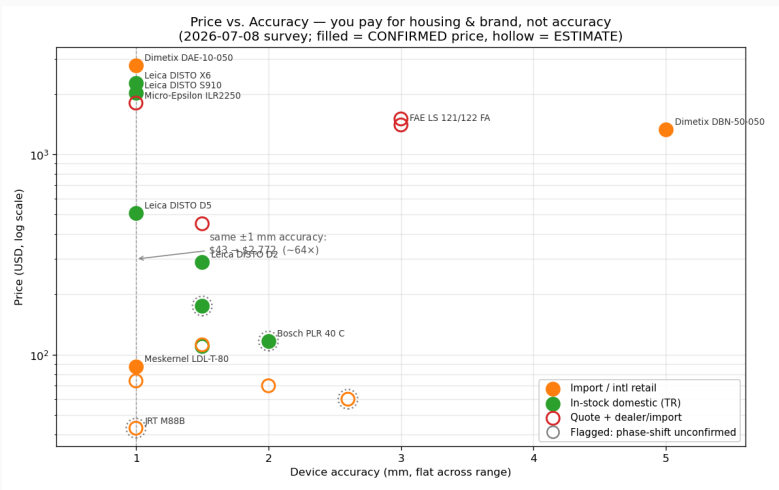
Device Shortlist (meets ≤ 40 m / ≤ 10 mm)

Device	Range	Accuracy	Interface	Price
Meskernel LDL-T	0.03–80 m	± 1 mm, 100 Hz	RS232/485/USART	\$87.50 c.
Dimetix DBN-50-050	0.05–50 m	± 5 mm, pub. ASCII	RS232 / RS422	\$1,332 c.
Leica DISTO D5	0.05–200 m	± 1.0 mm	BLE 5.0 + USB-C	\$311–708 c.
Leica DISTO S910	0.05–300 m	± 1.0 mm	BLE + native USB	\$1,916–2,132 c.
Bosch PLR 40 C [†]	0.05–40 m	± 2 mm	BLE 4.2	\$110–125
JRT B605B [†]	0.03–150 m	$\approx \pm 2.6$ mm @ 40 m	RS232 TTL	\$43–90

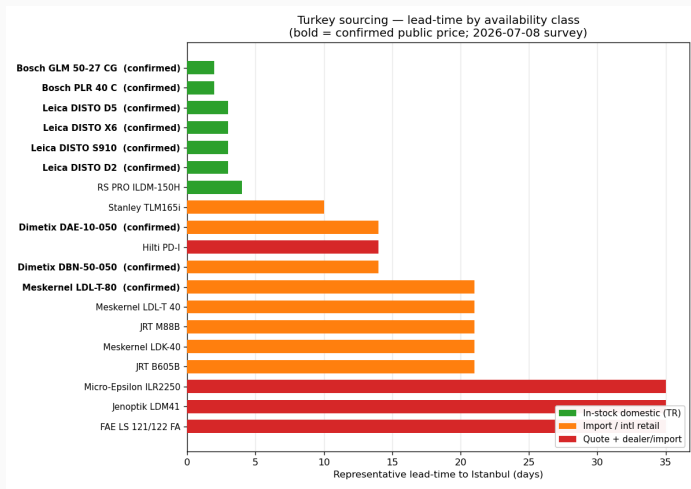
c. = confirmed price (2026-07-08 survey). [†] = phase-shift principle **unconfirmed** — held out of the certified list.

20 qualifying/flagged devices total. Accuracy is never the problem; **price, sourcing, and interface** are the real decisions.

Price vs. Accuracy — You Pay for Housing, Not Accuracy



Turkey Sourcing — Lead-Time by Availability Class



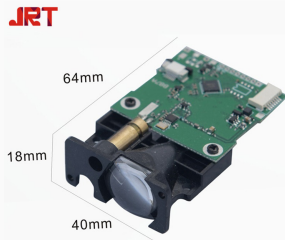
Flagged: Attractive, but Principle Unconfirmed

The two cheapest/fastest options are **held out of the certified list**: the 2026-07-08 survey could not confirm their measurement principle is phase-shift from any primary source. Verify before committing.



Bosch PLR 40 C[†]

±2 mm, best Turkey availability, BLE —
principle unconfirmed.



JRT B605B[†]

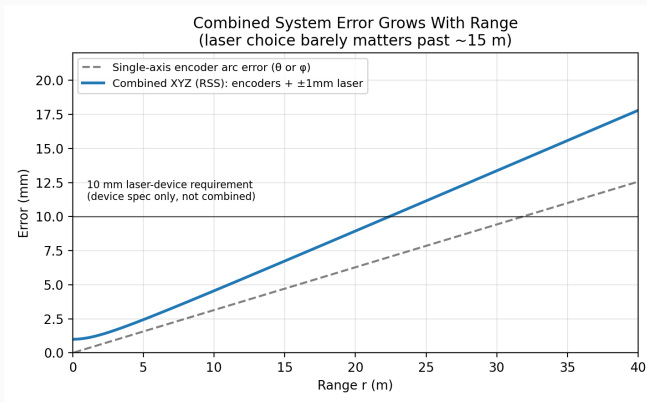
Cheapest wired (\$43), docked (A) or fixed (B)
— **principle unconfirmed.**

Recommendation — Bench-Test Order (confirmed-phase-shift first)

1. **Meskernel LDL-T** — order first. Cheapest device with confirmed phase-shift *and* confirmed price (\$87.50); ± 1 mm, 100 Hz. Request the command datasheet before ordering.
2. **Dimetix DBN-50-050** (\$1,332) — the only **public ASCII protocol** in the study; de-risks the firmware parsing path. DAE-10-050 (\$2,700) for ± 1 mm.
3. **Leica DISTO D5** (\$311–708, in TR) — confirmed-phase-shift ± 1 mm handheld reference; clean BLE-path validation.
4. **Flagged Bosch / JRT / RS PRO** — cheap & fast, but only *after* a datasheet confirms phase-shift. Industrial tier (ME / WayCon / FAE) only if certified housing is required.

Context: System Error Budget (out of scope)

The ≤ 10 mm spec is the *laser device* alone. The encoders' angular error **grows with range** — past ~ 15 m they dominate and **laser choice stops mattering** to total accuracy (≈ 18 mm at 40 m).



Conclusion

The ≤ 40 m / ≤ 10 mm requirement is comfortably met —
but only with phase-shift lasers.

A cheap bench path (Bosch + RS PRO + JRT) validates
the integration before spending on any industrial-grade unit.